

The self-auditing risk layer for Alpaca options.

An autonomous agent that trades defined-risk spreads through Alpaca's CLI — and re-prices the trades it refused, to prove the refusals were right.

20,000-path Monte Carlo VaR · Counterfactual refusal scoring

ACCOUNT

PA3BY0HJORNC

fresh, \$100,000, options level 3

EQUITY

\$100,149.93

live paper trading

EXECUTION

Alpaca CLI

multi-leg, on a schedule

THE PROBLEM

Risk paralysis.

Give a language model a brokerage API and it finds a reason to trade. Put a filter in front of it and it finds a reason to never trade.

Both failures look identical from the outside — a flat equity curve and a confident explanation. Nobody can tell an agent with discipline from an agent that is simply broken, because nobody measures the trades that were *not* taken.

UNCALIBRATED FILTER

0 trades

looks like discipline

NO FILTER

blown account

looks like conviction

WHAT'S MISSING

the counterfactual

what the refusals would have done

Every refusal becomes a testable claim.

01

Refuse

The gate vetoes a spread and logs its legs, credit and reason to an append-only ledger.

02

Re-price

Those exact legs are re-priced later at live marks — what the refused trade would have earned.

03

Grade

Scored symmetrically. A refusal that avoided a loss counts; so does one that missed a profit.

04

Recalibrate

The threshold moves on the evidence, and the scorer keeps grading the new one.

Before execution, a 20,000-path Monte Carlo prices the distribution — probability of profit, 95% VaR, expected shortfall. A trade with negative expected value is refused however much premium it appears to pay.

Predicted before the trade. Verifiable after it.

	MONTE CARLO, BEFORE	LIVE ALPACA EXECUTION
P&L on the 17-lot	+\$145 expected	+\$153 realised
Probability of profit	92%	won
95% VaR	-\$1,428	not breached
Structural max loss	-\$1,428	0 of 20,000 paths exceeded

open cd44d308-8609-4044-8994-3137d255fc78 17 @
-0.16 2026-09-01T17:10:59Z
close f67cd64a-a82c-4db8-a877-52de37037761 17 @
+0.07 2026-09-02T14:57:24Z

And the honest half: all four gate vetoes would have been profitable. We published that unchanged and lowered the floor on the evidence.

Alpaca options production hardening.

Four behaviours met on the live API, each with its receipt and the code that handles it. Two are documented and still cost a day.

One is not. One earlier claim is withdrawn.

- **Position intent is inferred from the book; a mismatch is a 422.** Four approved entries refused on 3 Sep, request ids in the post-mortem. Dry run cannot see it. Fixed with a book pre-flight and ten tests.
- **Multi-leg limit sign is documented and still bites.** The CLI help doesn't say; the first order went in with the wrong sign and filled anyway. Now refused in code.
- **Greeks on a paper key return null, not an error.** The default feed is the one a paper key can't read. `feed=indicative` .
Reproduced 4 Sep. Always
- **Withdrawn:** the "\$100,000 intraday offset" did not reproduce and has no ledger record.

MIT licensed. The workaround layer is the contribution: it turns Alpaca's options API from something you discover by trial into something you build on.